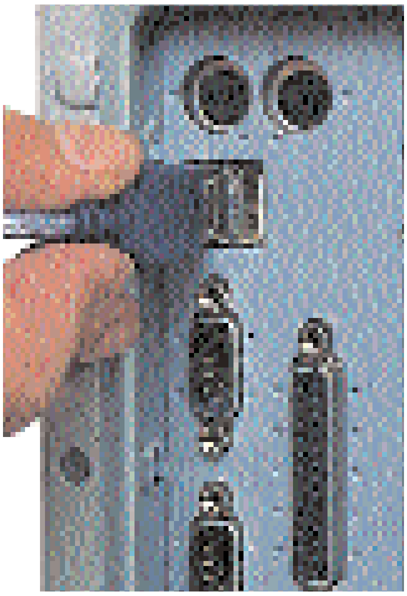
Operating systems

The code that runs our desktops, servers and operating systems has seen a visual metamorphosis over the past year, with increasin stress on better looking and more intuitive interfaces. Things have changed under the skin too, with stronger protection against external threats, faster networking and support for new hardware. Through it all, the familiar 'desktop' metaphor has remained with the folders, recycle bin and wallpaper.

While there is no sign of breaking away from this metaphor, upcoming OSes will offer greater levels of stability and interfaces. They will also serve as a platform for unifying new-age applications such audio and video editing, better media file management and a better broadband experience. OSes will also move into the 64-bit realm to complement the arrival of 64-bit processors and applications.

On the desktop front, Microsoft's new OS, codenamed Longhorn, boasts of a brand new interface, a powerful new database-like file system, support for Palladium technology, as well as provision for 3D hardware for a more realistic and visually rich interface. On the Linux front, the launch of United Linux assures compatibility and higher ease of use. Other distributions of Linux will also focus on the end user where user-friendliness will be enhanced. The OSes for mobile devices will also change to offer stronger multimedia support—the upcoming versions of Microsoft Pocket PC 2002 and Palm OS 5 support faster processors, full audio and video.



Connectivity

Apart from the processor and the motherboard, peripherals and other system components are also getting faster and are demanding greater levels of bandwidth from interfaces—both internal and external. While the year gone by has seen advancements in the speed of interfaces such as ATA and USB, this trend will continue with greater performance through innovations in the design itself.

Serial ATA is one such standard that will make a big appearance this year—with a 150 MBps bandwidth, this interface is faster than any existing IDE interface we have today. To corroborate this fact, there is no future roadmap charted out for the parallel ATA standard. Since Serial ATA uses thinner connecting cables, there will be far less clutter in your system with greater cooling and therefore, more stable performance. Several hard disk manufactures have already launched hard disk drives based upon this standard.

On the external front, USB 2.0 took time to gain acceptance during the beginning of 2002. Compared to USB 1.1's paltry 12 Mbps bandwidth, USB 2.0 features a bandwidth of 480 Mbps allowing for fluid operation of devices such as DV cameras, external storage devices, scanners, etc. With strong support from manufacturers of motherboards as well as peripheral devices, you can look forward to the USB 2.0 interface peripheral devices.

The visual realm

Advancements in graphics processing hardware have made cartoon-like game characters a thing of the past. The buzzword today is visual computing and this is apparent from the current generation of games, conventional application software and operating systems. With today's OSes drawing on the capabilities of the graphics processor for even rendering a shaded and bevelled scroll bar, our entire experience with computers is moving closer towards a visual realm.

In the race for launching the fastest graphics card, the two bigwigs in graphics processors, ATi and nVidia, are going all out on the promotions and enhancements. This is supplemented by the fact that even upcoming graphics APIs such as DirectX 9 and OpenGL 2.0 boast of capabilities that will bring cinema-quality graphics to your computer. The year 2002 has already given us a taste of things to come with ATi's Radeon 9700. This year, we will witness some more powerful graphics solutions with products such as the GeForce FX and ATI's upcoming R10000 graphics processor.